

AI-powered drone inspection for critical energy infrastructure, leveraging acquired Cyberhawk capabilities.

Energy × Critical infrastructure monitoring × Drones and UAS · OS113 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
73 is the world moving	54 can we play, can we win	MEDIUM 0 named in the evidence	€7.8m partial evidence · per year	NEXT pilots published no volume pr...	L0 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is an AI-powered drone inspection service for critical energy infrastructure, leveraging Orange's Drone Guardian offer and IoT expertise. It targets energy operators who need to monitor assets like power lines, solar farms, and pipelines more safely and efficiently. The market is moving now because of recent acquisitions in AI-powered infrastructure intelligence and growing concerns about infrastructure security and counter-drone gaps.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€9.5m €1.5m – €146m	€7.8m €1.2m – €120m	€117k €18k – €1.8m	partial
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.6m €2.6m – €6.9m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Drones and UAS	2.4 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TAR	2025
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Artificial intelligence by NACE Rev. 2 activity series E_AI_TAR is used as a proxy for Drones and UAS: Autonomous machines — includes autonomous drones. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (UN); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

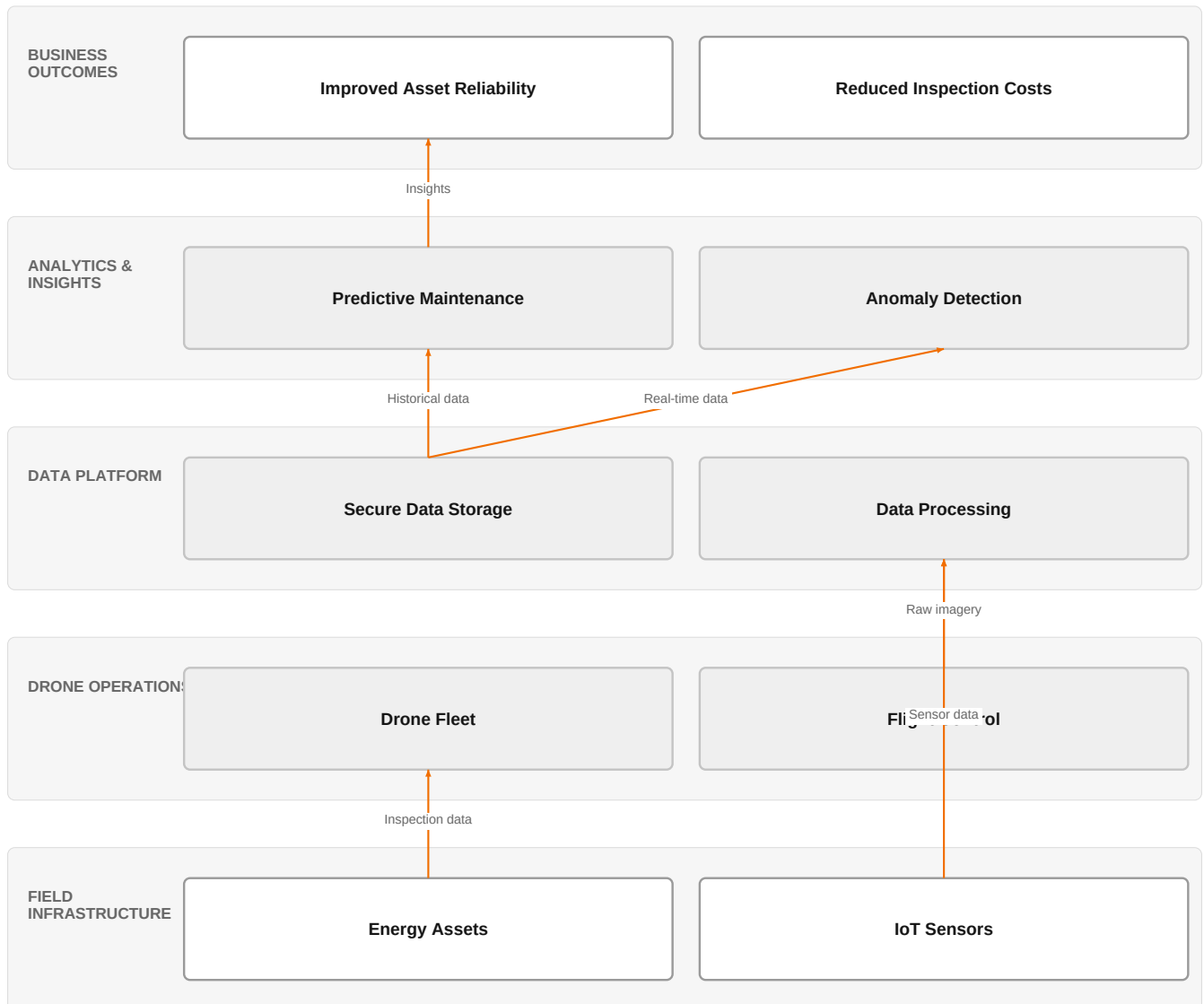
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (UN); the estimate covers the European footprint only.

The solution, in one picture

AI-Powered Drone Inspection for Energy Infrastructure



This diagram shows how Orange's secure data platform and analytics turn drone and IoT data into actionable insights for energy infrastructure operators.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

Who buys, and why now

The buyer is typically the head of asset management or operations at an energy utility, who feels the pain of manual inspections that are slow, costly, and sometimes dangerous. The trigger is often a regulatory audit, a major incident, or a board-level directive to improve resilience and reduce downtime. They need to inspect vast infrastructure like transmission lines or solar farms more frequently and with better data, without putting workers at risk.

Why it is hot now — every claim bound to a dated source

- Ondas completed the acquisition of Cyberhawk, an AI-powered critical infrastructure intelligence leader, indicating market momentum. [SIG-8588B752FF10]
- Ondas completed the acquisition of Cyberhawk, a leader in AI-powered critical infrastructure intelligence. [SIG-72531479498D, SIG-8588B752FF10, SIG-BB3788AF4131]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a drone inspection service using the Orange Drone Guardian offer, which provides secure drone operations and data management. This would be complemented by Orange's IoT experts to integrate sensor data and Orange Group researchers for advanced analytics. The service would be delivered on Orange's certified infrastructure (ISO 27001, SecNumCloud) to ensure data sovereignty and security. The engagement would start with a pilot on a specific asset type, then scale to full operations.

Why Orange can win

Orange can win because it combines a certified, sovereign cloud infrastructure (SecNumCloud, ISO 27001) with a dedicated drone offer (Orange Drone Guardian) and deep IoT expertise. This is a unique combination that addresses both the operational need and the security and compliance requirements of critical infrastructure operators. The assets are strong, but the offer is still evolving, so we must position it as a trusted, end-to-end solution.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Drone Guardian	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 5.7; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include security specialists like Airbus Protect, Dedrone by Axon, HENSOLDT, Leonardo, and Thales, who sell drone and counter-drone systems. Global systems integrator Capgemini is active in energy, and telecom peer Telefonica Tech also targets energy. Honeywell brings industrial OT expertise. These competitors are strong in hardware and specific security niches, but Orange differentiates with its sovereign cloud and integrated telecom capabilities.

Sources: SIG-37D6B973F20E bing.com 2026-08-12; SIG-BCBC462A63E3 Police Magazine 2026-07-09

Competitor	Category	Position here	Basis	Evidence
Airbus Protect	Security specialist	sells Drones and UAS	structural	—
Dedrone by Axon	Security specialist	sells Drones and UAS; competes in OX: Smart Industries	structural	—
HENSOLDT	Security specialist	sells Drones and UAS	structural	—
Leonardo	Security specialist	sells Drones and UAS	structural	—
Thales	Security specialist	sells Drones and UAS	structural	—
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—
Telefonica Tech	Telecom operator peer	active in Energy	structural	—
Honeywell	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently inspect your critical energy infrastructure, and what are the main pain points in terms of cost, safety, or frequency?
- 2 Have you experienced any recent incidents or near-misses that highlighted the need for more frequent or safer inspections?

- 3 What is your current approach to data security and sovereignty for operational data collected from drones or IoT sensors?
- 4 Are you under any regulatory pressure to improve infrastructure monitoring or reporting?
- 5 Have you evaluated any drone-based inspection solutions before? If so, what were the main barriers to adoption?

Objections you will meet

They will say	You can answer
We already have our own drone inspection program.	That's great, and it shows you understand the value. Our solution can complement your existing program by providing a sovereign, secure data platform and advanced analytics that you may not have in-house. We can also help you scale to more assets without significant capital investment.
Drone inspections are not reliable enough for our critical infrastructure.	Reliability is a valid concern. Our solution leverages AI and IoT expertise to improve detection accuracy, and we can run a pilot on a non-critical asset to demonstrate the quality of data and insights before you commit to broader deployment.
We are concerned about data security and regulatory compliance.	That's exactly why we built our solution on SecNumCloud and ISO 27001 certified infrastructure. We ensure that all data is processed and stored in sovereign environments, meeting the highest standards for critical infrastructure. We can walk you through our compliance framework in detail.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: for a European energy transmission operator's tender on drone-based infrastructure inspection, draft a proposal that highlights Orange Drone Guardian's secure data handling (ISO 27001, SecNumCloud) and how our IoT experts and Orange Group researchers can tailor AI analytics to their asset base.
Sales	In a meeting with a European energy transmission operator's head of asset management, say: 'Orange Drone Guardian, backed by our ISO 27001 and SecNumCloud certifications, can give you a secure, end-to-end drone inspection service—let's explore how it fits your critical infrastructure monitoring needs.'
Strategist / Innovator	Deep-dive brief: assess how Orange Drone Guardian can be extended with Cyberhawk-style AI inspection analytics, and validate the drone-inspection value chain with two European energy transmission operators.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-0CB746316F44	2026-05-21	International Journal of ...	1	Industrial Control System Cyber Security Frameworks: Assessment using Defense-in-Depth Criteria
SIG-CD11CC6D802B	2026-04-26	National Education Servic...	1	Photonic–Tensor AI for Ultra-Low-Latency Edge–Cloud Computing
SIG-0860D0B1EBA0	2025-12-05	openalex.org	1	Deploying TinyML for energy-efficient object detection and communication in low-power edge AI s...
SIG-39681DB73044	2025-09-01	openalex.org	1	AI-Integrated autonomous robotics for solar panel cleaning and predictive maintenance using dro...
SIG-B7834050E076	2025-08-26	openalex.org	1	A Case Study of Environmental Footprints for Generative AI Inference: Cloud versus Edge
SIG-950CF0EDAE0F	2025-08-18	openalex.org	1	Secure Communication in Drone Networks: A Comprehensive Survey of Lightweight Encryption and Ke...
SIG-37D6B973F20E	2026-08-12	bing.com	2	Lockheed Martin demonstrates AI-powered drone detection system using 5G networks
SIG-72531479498D	2026-08-10	finanznachrichten.de	2	Ondas Inc .: Ondas Completes Previously Announced Acquisition of Cyberhawk , a Leader in AI - P...
SIG-8588B752FF10	2026-08-10	californiatelegraph.c om	2	Ondas Completes Previously Announced Acquisition of Cyberhawk , a Leader in AI - Powered Critic...
SIG-BB3788AF4131	2026-08-10	itnewsonline.com	2	IT News Online - Ondas Completes Previously Announced Acquisition of Cyberhawk , a Leader in AI...
SIG-BCBC462A63E3	2026-07-09	Police Magazine	2	New DroneShield Report Reveals Serious Gaps in Airport, Critical Infrastructure Counter-Drone S...
SIG-DEDC91C97937	2026-08-01	arxiv.org	3	Collaborative Orbital Edge Intelligence: A Decentralized Paradigm for Energy-Efficient Computin...
SIG-D29A41589B0F	2026-07-25	arxiv.org	3	Movable-Antenna Assisted Energy Minimization in UAV-Enabled Mobile Edge Computing Systems
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomaly Detection in ...
SIG-17DA3B724138	2026-04-27	arxiv.org	3	System-aware contextual digital twin for ICS anomaly diagnosis

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: what_is_changing (named entities that were never supplied: Cyberhawk, Ondas); risks_and_unknowns (named entities that were never supplied: Cyberhawk, Ondas); diagram (5 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:10+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.